

Mayank Goel

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EDUCATION

International Institute of Information Technology - Hyderabad

B.Tech in Computer Science, CGPA: 7.96/10

Hyderabad, India

Jul. 2019 – Apr. 2023

International Institute of Information Technology - Hyderabad

MS in Computational Linguistics, CGPA: 7.92/10

Hyderabad, India

May 2023 – Dec. 2024

EXPERIENCE

Research Engineer

H2Loop.ai

Apr. 2026 - Present

Bangalore, India

- Implementing recipes for data generation and environments for SFT/RL. Finetuning LLMs on these tasks for significant performance gains on low-level languages on agentic coding. Owning the entire end-to-end model finetuning pipeline including creating robust evaluations.

Researcher

Lossfunk

Aug. 2025 - Mar. 2026

Bangalore, India

- Exploring new paradigms to approach AI Safety. Primary topics of interest were moral philosophy, game theory, multi-agent systems.
- An exploration of how models think about trolley problems through mechanistic and empirical means through a custom architecture was accepted at the Moral Ethics Workshop, AAI '26

Research Mentor

Algoverse Research

Jul. 2025 - Dec. 2025

Remote

- Mentoring teams as part of a research bootcamp, consisting primarily of college students and early-stage engineers. Responsibilities include holding meets, discussing and coming up with the direction, unblocking team members when needed, and acting as liaison to senior mentors and organizers.
- Works from this have been accepted to the SEA Workshop, NeurIPS '25, ICLR '26 Main, MALGAI Workshop, ICLR '26 .
- Executed skills about effective conflict management while maintaining strong epistemic hygiene.

Research Engineer

IBM Research

Mar. 2025 - Jul. 2025

Bangalore, India

- Involved with synthetic data generation for SFT data, and environment building for RL, along with efforts on evals. Implemented several papers to help the data layer of the model training, with emphasis on synthetic math and coding data.
- Implemented a scalable data-rewriting pipeline using vLLM, building infrastructure for automated, high-throughput inference. Optimized GPU utilization across nodes by managing concurrency and resource sharing to minimize idle time and maximize throughput.

Research Engineer

Deccan AI

Sep. 2024 – Mar. 2025

Hyderabad, India

- Involved with benchmarking of models on various tasks, across modalities, such as tasks on image, speech, code. From searching for the appropriate metrics to explaining them to client-facing teams.
- Worked on automated QC for data generated by human annotators. Included fine-tuning models on curated data as part of a company repository.

Research Intern

ComplexData Lab, MILA AI

Oct. 2023 – Jul. 2024

Remote

- Designed experiments using the Debate framework for uncertainty quantification of LLM outputs.
- Worked on experiments to analyze and visualize data from other team projects on uncertainty quantification. Work done during this phase was accepted to COLM '24 and NeurIPS SafeGenAI Workshop '24

Research Intern

Robert Bosch GmbH

Jun. 2023 – Jul. 2023

Bangalore, India

- Extended an hierarchical clustering algorithm to use sentence embeddings. Built a pipeline for domain adaptation of sentence embeddings using various training methods such as denoising, synthetic QnA generation. Ran evals across key metrics in clustering, document retrieval.

- Involved with LORA Finetuning of LLMs on various domains, along with setting up appropriate evaluation criteria.
- Developed an e-commerce website crawling pipeline using Apache Nutch and Selenium, with a pipeline to go from seed keywords to a set of relevant web documents.
- Built a Spacy3 model utilizing Transformers for a production-grade custom NER model for an e-commerce company, as part of an end-to-end system. Generated NER data automatically by fuzzy matching table rows on item descriptions. Achieved accuracy of 91% on a multi-class system, with various additional heuristics to help model performance.

PUBLICATIONS

Interpretable Multi-Agent Debate for Political Opinion Simulation Multi-Agent Learning Workshop, ICLR '26

Authors: Aali Azamat Uulu, Justin Xue Taing, Alibek Dadajonov, **Mayank Goel**

Automated Stateful Specialization for Adaptive Agent Systems ICLR '26

Authors: Myan Vu, Harrish Ayyanar, Pang Jiang, Anwiketh Reddy, **Mayank Goel**

Building Interpretable Models for Moral Decision-Making AAAI '26, Machine Ethics Workshop

Authors: **Mayank Goel**, Aritra Das, Paras Chopra

Automated Specialization of Stateful Agent Systems NeurIPS '25 Scaling Environments for Agents Workshop

Authors: Myan Vu, Harrish Ayyanar, Pang Jiang, Anwiketh Reddy, **Mayank Goel**, Kevin Zhu

Conceptual Limitations of Current AI Safety Approaches and Virtue Ethics as an Alternative Technical AI Safety Conference '25

Authors: Masaharu Mizumoto, Rujuta Karekar, Mads Udengaard, **Mayank Goel**, Daan Henselmans, Nurshafira Noh, Saptadip Saha, Pranshul Bohra

Epistemic Integrity in Large Language Models NeurIPS '24 SafeGenAI Workshop

Authors: Bijean Ghafouri, Shahrard Mohammadzadeh, James Zhou, Pratheeksha Nair, Jacob-Junqi Tian, **Mayank Goel**, Reihaneh Rabbany, Jean-François Godbout, Kellin Pelrine

Web Retrieval Agents for Evidence-Based Misinformation Detection COLM '24

Authors: Jacob-Junqi Tian, Hao Yu, Yury Orlovskiy, Tyler Vergho, Mauricio Rivera, **Mayank Goel**, Zachary Yang, Jean-François Godbout, Reihaneh Rabbany, Kellin Pelrine

Joke Generation Using Masked Language Infilling on Automatically Extracted Templates Humor Research Conference '23

Authors: **Mayank Goel**, Abhijit Manatkar

Towards conversational humor analysis and design Humor Research Conference '21

Authors: Tanishq Chaudhary*, **Mayank Goel***, Radhika Mamidi

Automated Evaluation of Conversational Prompts using Ordinal Ranking for Humor Generation ICON '24

Authors: **Mayank Goel**, Parameswari Krishnamurthy, Radhika Mamidi

TECHNICAL SKILLS

Languages: Python, Bash

Frameworks: PyTorch, HuggingFace Transformers, Streamlit, Selenium, SpaCy

AWARDS AND OTHER RELEVANT INFORMATION

- **Research Award** from IIIT Hyderabad in 2021
- **Top 30** in the **Indian** edition of **International Olympiad of Linguistics**
- **Fellow, ARENA 2024, Spring**
- **Participant, AI Safety Fundamentals Course, BlueDot Impact**
- **Teaching Assistant** for **Introduction to NLP** in Spring '22, **Advanced NLP** in Monsoon '23
- **Ex-head** of University **Debate Society** and **Humor Club**
- **Founder-Moderator** of **Philosophy Friday Discussion Group**
- **Best Project, Theme Track** in **Natural Language API Hackathon** by **expert.ai**
- **1st Prize, Equity Track** in **HackUMBC**, University of Maryland
- **Best Web Application** in **HackAtHome**, Brown University
- **3rd Prize, Walmart Track** in **TAMU Datathon**, Texas A&M University
- **1st Prize, Stellantis Track** in **Megathon '22**, IIIT Hyderabad